Wind effect on **wildfire** and **simulation** of its **spread** (case study: Siahkal forest in northern Iran).

R Jahdi, AA Darvishsefat, V Etemad, MA Mostafavi - 2014 - researchgate.net

… **Wildfire spread** and **behavior prediction** is important to bring the fire under control. This study validated the use of the FARSITE **model** in Northern Forests of Iran. The direction of **spread** …

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Information systems in support of **wildland fire management** decision making in Canada

BS Lee, ME Alexander, BC Hawkes, TJ Lynham… - … and Electronics in …, 2002 - Elsevier

… on the CFFDRS as a means of predicting **wildland fire** occurrence and **behavior** potential. … These analyses can be used to support the setting or **review** of policy, allocation of national or …

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Effects of bark beetle-caused tree mortality on **wildfire**

JA Hicke, MC Johnson, JL Hayes… - … Ecology and **Management**, 2012 - Elsevier

… and crown fire **behavior** responses, leading to … **wildfire** such as climate. However, some disagreement among studies remained. Given the large areas of recent bark beetle and **wildfire** …

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Wind–terrain effects on the **propagation** of wildfires in rugged terrain: fire channelling

JJ Sharples, RHD McRae… - … Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… Implications of fire channelling for **bushfire** risk **management** are also discussed. … of the atypical fire **spread** as ‘fire-channelling’ events. By applying a simple terrain-filter **model** to the …

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Fuels and fire **behavior dynamics** in bark beetle-attacked forests in Western North America and implications for fire **management**

MJ Jenkins, WG Page, EG Hebertson… - … and **Management**, 2012 - Elsevier

… in forest health and **wildfire** protection policies of land **management** agencies. … **management** and threaten the safety of firefighters and the general public alike. In this paper, we **review** …

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Predicting the Future of Forest Fires: Climate Models, Fire **Behavior** and Fire **Spread**

P Mishra, H Singh, N Kumar - **Forest Fire** and Climate Change: Insights …, 2025 - Springer

… These models allow for the hourly **prediction** of **forest fire spread** after identifying the ignition … The **review** has flaws in everything from the coordination of preventative measures to the …

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A **wildfire** risk assessment framework for land and resource **management**

JH Scott, MP Thompson, DE Calkin - 2013 - digitalcommons.unl.edu

… , geospatial information regarding the likely occurrence, **spread**, and consequences of **wildfire**. The use of burn probability … One such **survey** used in the past queried experts regard- …

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How forest fires kill trees: a **review** of the fundamental biophysical processes

ST Michaletz, EA Johnson - Scandinavian Journal of Forest …, 2007 - Taylor & Francis

… Firsts, a more sophisticated fire **behavior model** is required so that the flaming fire front can … us to develop and improve our understanding of **forest fire behavior** and effects. K. Miyanishi, …

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Charts for interpreting **wildland fire behavior** characteristics

PL Andrews, RC Rothermel - 1982 - books.google.com

… As an illustration of how the fire **behavior** chart works, the fire **behavior prediction** values … A **review** of each index and the relationship between them will clarify the use of the NFDRS …

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AEGIS: a **wildfire** prevention and **management** information system

K Kalabokidis, A Ager, M Finney… - … Hazards and Earth …, 2016 - nhess.copernicus.org

… fire exposure to highly valued resources, and to **model** the effect of fuel breaks on **wildfire spread**. Under constant … and Ager, AA: A **review** of recent advances in risk analysis for **wildfire** …

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Setting **wildfire** evacuation triggers by coupling fire and traffic **simulation** models: a spatiotemporal GIS approach

D Li, TJ Cova, PE Dennison - Fire technology, 2019 - Springer

… The results in this study demonstrate that we could better reveal the **dynamics** of evacuation traffic and fire **spread** in **wildfire** evacuations when we couple the two models to set triggers. …

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A **review** of the applications of remote sensing in fire ecology

DM Szpakowski, JLR Jensen - Remote sensing, 2019 - mdpi.com

… **wildfire spread**, wind direction, and weather conditions. An example of a commonly used **model** … FARSITE then uses existing fire **behavior** models to simulate how the fire will grow and …

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Deep learning and transformer approaches for UAV-based **wildfire** detection and segmentation

R Ghali, MA Akhloufi, WS Mseddi - Sensors, 2022 - mdpi.com

… Early detection and **prediction** of fire **spread** can help reduce affected areas and improve firefighting. Numerous systems were developed to detect fire. Recently, Unmanned Aerial …

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Self-organized critical **forest-fire model**

B Drossel, F Schwabl - Physical **review** letters, 1992 - APS

… , the **forest-fire** … the **model** rules, eg, anotherlattice symmetry or fire spreading to next-nearest neighbors. Let us compare the critical **behavior** of the **forest-fire model** to the sandpile **model**…

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Incidence of **forest fire** in India and its effect on terrestrial ecosystem **dynamics**, nutrient and microbial status of soil

KK Chandra, AK Bhardwaj - International Journal of Agriculture and …, 2015 - academia.edu

… **management** of forest after fire will be more beneficial. The present **review** deals the impact of **forest fire** on species **dynamics**, … **management** and control of nutrient, insect and disease. …

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A **review** of initial attack fire crew productivity and effectiveness

KG Hirsch, DL Martell - International Journal of **Wildland Fire**, 1996 - CSIRO Publishing

… and **spread** throughout the **forest fire management** community during the past 30 years. This information has been incorporated into a wide range of fire **management** … we **review** several …

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A real-time risk assessment tool supporting **wildland fire** decisionmaking

DE Calkin, MP Thompson, MA Finney… - Journal of …, 2011 - academic.oup.com

… fire **behavior** and **wildfire** impact modules of the **Wildland Fire** … fire **model** components, Fire **Spread** Probability (FSPro; fire … We **review** some real-world applications of WFDSS and show …

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R Lovreglio, O Meddour-Sahar… - Iforest-Biogeosciences and …, 2014 - iforest.sisef.org

… to change **wildfire behavior**, by slowing its **spread**, shortening … The pastoralism, partner of **forest fire management**: 20 years … Europe, Forest **management** as a tool for **wildfire** prevention”. …

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Rethinking the **wildland fire management** system

MP Thompson, DG MacGregor, CJ Dunn… - Journal of …, 2018 - academic.oup.com

… of **behavior**, and in so doing, challenge ingrained ways of thinking and acting that may be inconsistent with current intentions around **wildland fire management**. We … Such a **review** could …

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Time aware genetic algorithm for **forest fire propagation prediction**: exploiting multi-core platforms

T Artés, A Cencerrado, A Cortés… - Concurrency and …, 2017 - Wiley Online Library

… **forest fire** scenario and accomplishing it within strict time requirements to deliver a **forecast**, in this work, a complete **forest fire spread prediction** … accurate **propagation prediction** results …

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